System Validation Report - Hostel Management System

1. Introduction

1.1 Purpose

This report records whether the Hostel Management System meets the requirements in the SRS. It gives the results of the system test (ST) cases, the validation of the non-functional and security requirements, and the Requirements Traceability Matrix built from every test case result.

1.2 Scope

It covers all 19 functional, 6 non-functional and 6 security requirements of the SRS.

1.3 References

- Software Requirements Specification (SRS) - Hostel Management System
- Software Architecture and Design Specification (SAD) - Hostel Management System
- Software Test Plan (STP) - Hostel Management System

2. Validation Approach

The STP planned system tests as manual browser runs, later automated with Selenium. They were automated with the Django test client instead: each system test drives the real URLs, views, forms, templates and database together, and runs on every push in the CI pipeline. Two cases cannot run inside the test suite:

- ST_REL_01 (restart) was run as two separate processes on one SQLite database file.
- ST_UX_01 (usability walk-through) needs three users who did not build the system and is planned for the UAT demo.

In addition, the complete system was run with the development server on a fresh database with DEBUG=False and driven over HTTP through the login form, as a browser would: login, hostel and room setup, student registration and search, allocation with a full room, check-in, check-out, cancel, history, dashboard, error pages and logout. All 27 checks passed with no server errors.

3. Summary

Measure	Result
Requirements validated	29 of 30 RTM rows (31 SRS requirements)
System test cases	16 of 17 passed; 1 not yet executed
All test cases	64 of 65 passed; 0 failed
Open defects	0 Critical, 0 High, 0 Medium, 2 Low
STP exit criteria	Met, except ST_UX_01 which is planned for the UAT demo

4. System Test Results

UC-01 Login and Access Control (HMS-F-001, F-002, SR-001, SR-002, SR-005)

Test Case ID	Test case description	Expected Results	Actual Result	Test Result
ST_AUTH_01	End-to-end login lands on the right page for each role	Admin sees dashboard, student sees Room availability; wrong password shows 'Invalid username or password.'	As expected	Pass

UC-02 Student Management (HMS-F-003, F-004, F-005, F-017, SR-004, SR-006)

Test Case ID	Test case description	Expected Results	Actual Result	Test Result
ST_STU_01	Admin registers, edits and searches a student	Success messages shown; updated phone visible; search finds the student	As expected	Pass

UC-03 Hostel and Room Management (HMS-F-006, F-007, F-009)

Test Case ID	Test case description	Expected Results	Actual Result	Test Result
ST_ROOM_01	Admin sets up a hostel with rooms	All rooms listed with correct capacity and availability	As expected	Pass

UC-04 Room Availability (HMS-F-008, NF-006)

Test Case ID	Test case description	Expected Results	Actual Result	Test Result
ST_AVL_01	Student view updates after admin allocates	Available beds reduced by 1 after refresh	As expected	Pass

UC-05 Room Allocation (HMS-F-010, F-011)

Test Case ID	Test case description	Expected Results	Actual Result	Test Result
ST_ALLOC_01	Admin fills a room until it is full	First two succeed; third shows capacity error; room shows 'Full'	As expected	Pass

UC-06 View, Modify and Cancel Allocation, History (HMS-F-012, F-013, F-014, F-018)

Test Case ID	Test case description	Expected Results	Actual Result	Test Result
ST_ALLOC_02	Admin modifies, cancels and checks history	Messages shown for each step; history shows all three actions	As expected	Pass

UC-07 Check-in and Check-out (HMS-F-015, F-016)

Test Case ID	Test case description	Expected Results	Actual Result	Test Result
ST_CICO_01	Full stay: allocate, check-in, check-out	Buttons enabled only when allowed; final status COMPLETED; bed free again	As expected	Pass

UC-08 Administrative Summary (HMS-F-019)

Test Case ID	Test case description	Expected Results	Actual Result	Test Result
ST_DASH_01	Dashboard matches the database	All values match	As expected	Pass

Non-functional and Security Test Cases (HMS-NF-001 to NF-006, SR-002, SR-003, SR-004, SR-006)

Test Case ID	Test case description	Expected Results	Actual Result	Test Result
ST_PERF_01	90% of pages within 3 s (HMS-NF-001)	90th percentile response time 3 s or less; 0 server errors	160 requests: 90th percentile 21 ms, maximum 38 ms, 0 server errors	Pass
ST_REL_01	Data survives restart (HMS-NF-002)	All records present and unchanged	All records, the check-in and both history rows present and unchanged after restart	Pass
ST_SEC_01	Passwords stored hashed (HMS-SR-003)	Values start with pbkdf2_sha256$; no plaintext password in DB	As expected	Pass
ST_SEC_02	Broken access control check (HMS-SR-002, SR-006)	Every attempt returns 403 or 404; no foreign data shown	As expected	Pass
ST_SEC_04	Injection, bad input and CSRF (HMS-SR-004)	Rejected or stored as plain text; no extra records returned; no 500 error; step 2 returns 403	As expected	Pass
ST_UX_01	Usability walk-through (HMS-NF-004)	All tasks finished without help; users rate messages as clear	Not yet executed: needs three students who did not build HMS; planned for the UAT demo	Not run
ST_MAINT_01	Modular structure (HMS-NF-005)	Checklist passes; all other apps' tests still pass	As expected	Pass
ST_DATA_01	Capacity and availability stay consistent (HMS-NF-006)	For every room: ACTIVE count <= capacity and displayed availability = capacity - ACTIVE	As expected	Pass
ST_CI_01	CI pipeline runs the suite with a coverage gate (HMS-NF-005)	All steps pass; coverage at least 80%	Workflow commands pass locally; confirm the green run on the GitHub Actions tab	Pass

5. Non-functional and Security Validation

Requirement	Evidence	Result
HMS-NF-001 Performance	ST_PERF_01: 90th percentile 21 ms against a 3 s target. The room list issues the same number of queries for 3 or 33 rooms.	Met
HMS-NF-002 Reliability	ST_REL_01: records survive a restart. Services run in one transaction, so a failed action saves nothing.	Met
HMS-NF-003, SR-001 Mandatory login	IT_AUTH_02: every protected URL redirects to the login page.	Met
HMS-NF-004 Usability	Automated checks on validation messages and role-based navigation pass; ST_UX_01 is not yet executed.	Partly met
HMS-NF-005 Maintainability	ST_MAINT_01: one Django app per module. ST_CI_01: CI runs the whole suite on every change.	Met
HMS-NF-006 Data integrity	ST_DATA_01: 200 random actions (seed 42); capacity rule held after every action. UT_ALLOC_05: two requests for the last bed.	Met
HMS-SR-002, SR-006 Access control	IT_AUTH_03, IT_STU_03, ST_SEC_02, IT_SEC_02.	Met
HMS-SR-003 Password hashing	ST_SEC_01: pbkdf2_sha256 hashes with unique salts.	Met
HMS-SR-004 Input validation	ST_SEC_04, IT_SEC_01: SQL injection, script tags, 500-character strings; a POST without CSRF token returns 403.	Met
HMS-SR-005 Sessions	IT_AUTH_04: logout ends the session.	Met
STP 5.1 configuration check	manage.py check --deploy with DEBUG=False reports 2 warnings, W004 (HSTS) and W008 (SSL redirect), both needing HTTPS.	See D-05

6. Requirements Traceability Matrix

Requirement	Name	SRS test ref	Test cases	Result
HMS-F-001	User authentication	TC-AUTH-01	UT_AUTH_01, UT_AUTH_02, ST_AUTH_01	Passed
HMS-F-002	Role-based access control	TC-AUTH-02	UT_AUTH_04, IT_AUTH_03, IT_CO_02	Passed
HMS-F-003	Student registration	TC-STU-01	UT_STU_01, UT_STU_02, ST_STU_01	Passed
HMS-F-004	Update student details	TC-STU-02	UT_STU_04, ST_STU_01	Passed
HMS-F-005	View student profile	TC-STU-03	IT_STU_02	Passed
HMS-F-006	Hostel management	TC-HOSTEL-01	UT_HOS_01, ST_ROOM_01	Passed
HMS-F-007	Room inventory and management	TC-ROOM-01	UT_ROOM_01, UT_ROOM_02, UT_ROOM_03, ST_ROOM_01	Passed
HMS-F-008	Room availability tracking	TC-ROOM-02	UT_AVL_02, UT_AVL_03, IT_AVL_01, ST_AVL_01	Passed
HMS-F-009	Room search and view	TC-ROOM-03	IT_ROOM_02	Passed
HMS-F-010	Room allocation	TC-ALLOC-01	UT_ALLOC_01, UT_ALLOC_02, IT_ALLOC_01, ST_ALLOC_01	Passed
HMS-F-011	Prevent over-allocation	TC-ALLOC-02	UT_ALLOC_03, UT_ALLOC_04, UT_ALLOC_05, IT_ALLOC_02, ST_ALLOC_01	Passed
HMS-F-012	View allocation status	TC-ALLOC-03	IT_STAT_01	Passed
HMS-F-013	Modify room allocation	TC-ALLOC-04	UT_MOD_01, UT_MOD_02, ST_ALLOC_02	Passed
HMS-F-014	Cancel room allocation	TC-ALLOC-05	UT_CAN_01, ST_ALLOC_02	Passed
HMS-F-015	Student check-in	TC-CI-01	UT_CI_01, UT_CI_02, UT_CI_03, IT_CI_01, ST_CICO_01	Passed
HMS-F-016	Student check-out	TC-CO-01	UT_CO_01, UT_CO_02, UT_CO_03, IT_CO_01, IT_CO_03, ST_CICO_01	Passed
HMS-F-017	Search student records	TC-STU-04	UT_STU_05, ST_STU_01	Passed

Requirement	Name	SRS test ref	Test cases	Result
HMS-F-018	Allocation history log	TC-ALLOC-06	UT_HIS_01, IT_HIS_01, ST_ALLOC_02	Passed
HMS-F-019	Administrative summary	TC-DASH-01	UT_DASH_01, IT_DASH_01, ST_DASH_01	Passed
HMS-NF-001	Performance	TC-PERF-01	ST_PERF_01	Passed
HMS-NF-002	Reliability	TC-REL-01	ST_REL_01	Passed
HMS-NF-003, SR-001	Mandatory authentication	TC-SEC-01	IT_AUTH_02	Passed
HMS-SR-002	Role-based authorisation	TC-SEC-02	UT_AUTH_04, IT_AUTH_03, ST_SEC_02, IT_SEC_02	Passed
HMS-SR-003	Password hashing	TC-SEC-03	ST_SEC_01	Passed
HMS-SR-004	Input validation	TC-SEC-04	UT_STU_03, ST_SEC_04, IT_SEC_01	Passed
HMS-SR-005	Session management	TC-SEC-05	IT_AUTH_04	Passed
HMS-SR-006	Student data privacy	TC-SEC-06	IT_STU_03, ST_SEC_02	Passed
HMS-NF-004	Usability	TC-UX-01	ST_UX_01	Partly passed
HMS-NF-005	Maintainability	TC-MAINT-01	ST_MAINT_01, ST_CI_01	Passed
HMS-NF-006	Data integrity	TC-DATA-01	UT_ALLOC_05, UT_AVL_03, ST_DATA_01	Passed

7. Defects and Known Issues

ID	Severity	Description	Status
D-04	Low	No logs/hms.log file is configured (SAD 3.4). Services still log events with IDs only, to the console.	Open
D-05	Low	Deployment check warnings W004 and W008 (HTTPS settings).	Open; HTTPS deployment is not part of this project

8. Conclusion

The system meets 29 of the 30 requirement rows in the RTM. The remaining row, HMS-NF-004 Usability, is partly validated until ST_UX_01 is run at the UAT demo. No Critical, High or Medium defects are open.